Storage limits

Cloud Volumes ONTAP 9.6 release notes

NetApp
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Table of Contents

Storage limits . . . 1
Storage limits in AWS . . . 1
Storage limits in Azure . . . 7
Storage limits in Google Cloud . . . 11

Storage limits

Storage limits in AWS

Cloud Volumes ONTAP has storage configuration limits to provide reliable operations. For best performance, do not configure your system at the maximum values.

Maximum system capacity by license

The maximum system capacity for a Cloud Volumes ONTAP system is determined by its license. The maximum system capacity includes disk-based storage plus object storage used for data tiering. NetApp doesn't support exceeding this limit.

For some HA configurations, disk limits prevent you from reaching the 368 TiB capacity limit by using disks alone. In those cases, you can reach the 368 TiB capacity limit by tiering inactive data to object storage. Refer to capacity and disk limits below for more details.

License	Maximum system capacity (disks + object storage)
Explore	2 TiB (data tiering is not supported with Explore)
Standard	10 TiB
Premium	368 TiB
BYOL	368 TiB per license

For HA, is the license capacity limit per node or for the entire HA pair?

The capacity limit is for the entire HA pair. It is not per node. For example, if you use the Premium license, you can have up to 368 TiB of capacity between both nodes.

For an HA system in AWS, does mirrored data count against the capacity limit?

No, it doesn't. Data in an AWS HA pair is synchronously mirrored between the nodes so that the data is available in the event of failure. For example, if you purchase an 8 TiB disk on node A, Cloud Manager also allocates an 8 TiB disk on node B that is used for mirrored data. While 16 TiB of capacity was provisioned, only 8 TiB counts against the license limit.

Disk and tiering limits by EC2 instance

Cloud Volumes ONTAP uses EBS volumes as disks, with a maximum disk size of 16 TiB. The sections below show disk and tiering limits by EC2 instance type because many EC2 instance types have different disk limits. Disk limits are also different between single node systems and HA pairs.

The disk limits below are specific to disks that contain user data. The limits do not include the boot disk and root disk.

Disk limits are shown by instance for Premium and BYOL licenses only because disk limits can't be reached with Explore or Standard licenses.

Single node with a Premium license

Instance type	Max disks per node	Max system capacity with disks alone	Max system capacity with disks and data tiering
c4.4xlarge	34	368 TiB	368 TiB
c4.8xlarge	34	368 TiB	368 TiB
c5.9xlarge	22	352 TiB	368 TiB
c5.18xlarge	22	352 TiB	368 TiB
c5d.4xlarge	22	352 TiB	368 TiB
c5d.9xlarge	22	352 TiB	368 TiB
c5d.18xlarge	22	352 TiB	368 TiB
m4.4xlarge	34	368 TiB	368 TiB
m5.4xlarge	22	352 TiB	368 TiB
m5d.8xlarge	22	352 TiB	368 TiB
r4.2xlarge	34	368 TiB	368 TiB
r5.2xlarge	22	352 TiB	368 TiB
r5d.2xlarge	22	352 TiB	368 TiB

Single node with one or more BYOL licenses

Instance type	Max disks per node	Max system capacity with one license		Max system capacity with multiple licenses	
		Disks alone	Disks + data tiering	Disks alone	Disks + data tiering
c4.4xlarge	34	368 TiB	368 TiB	544 TiB	368 TiB x each license
c4.8xlarge	34	368 TiB	368 TiB	544 TiB	368 TiB x each license
c5.9xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license
c5.18xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license
c5d.4xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license
c5d.9xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license

Instance type	Max disks per node	Max system capacity with one license		Max system capacity with multiple licenses	
c5d.18xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license
m4.xlarge	34	368 TiB	368 TiB	544 TiB	368 TiB x each license
m4.2xlarge	34	368 TiB	368 TiB	544 TiB	368 TiB x each license
m4.4xlarge	34	368 TiB	368 TiB	544 TiB	368 TiB x each license
m5.xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license
m5.2xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license
m5.4xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license
m5d.8xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license
r4.xlarge	34	368 TiB	368 TiB	544 TiB	368 TiB x each license
r4.2xlarge	34	368 TiB	368 TiB	544 TiB	368 TiB x each license
r5.xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license
r5.2xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license
r5d.2xlarge	22	352 TiB	368 TiB	352 TiB	368 TiB x each license

HA pairs with a Premium license

Instance type	Max disks per node	Max system capacity with disks alone	Max system capacity with disks and data tiering
c4.4xlarge	31	368 TiB	368 TiB
c4.8xlarge	31	368 TiB	368 TiB
c5.9xlarge	19	304 TiB	368 TiB
c5.18xlarge	19	304 TiB	368 TiB
c5d.4xlarge	19	304 TiB	368 TiB

Instance type	Max disks per node	Max system capacity with disks alone	Max system capacity with disks and data tiering
c5d.9xlarge	19	304 TiB	368 TiB
c5d.18xlarge	19	304 TiB	368 TiB
m4.4xlarge	31	368 TiB	368 TiB
m5.4xlarge	19	304 TiB	368 TiB
m5d.8xlarge	19	304 TiB	368 TiB
r4.2xlarge	31	368 TiB	368 TiB
r5.2xlarge	19	304 TiB	368 TiB
r5d.2xlarge	19	304 TiB	368 TiB

HA pairs with one or more BYOL licenses

Instance type	Max disks per node	Max system capacity with one license		Max system capacity with multiple licenses	
		Disks alone	Disks + data tiering	Disks alone	Disks + data tiering
c4.4xlarge	31	368 TiB	368 TiB	496 TiB	368 TiB x each license
c4.8xlarge	31	368 TiB	368 TiB	496 TiB	368 TiB x each license
c5.9xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
c5.18xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
c5d.4xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
c5d.9xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
c5d.18xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
m4.xlarge	31	368 TiB	368 TiB	496 TiB	368 TiB x each license
m4.2xlarge	31	368 TiB	368 TiB	496 TiB	368 TiB x each license
m4.4xlarge	31	368 TiB	368 TiB	496 TiB	368 TiB x each license

Instance type	Max disks per node	Max system capacity with one license		Max system capacity with multiple licenses	
m5.xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
m5.2xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
m5.4xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
m5d.8xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
r4.xlarge	31	368 TiB	368 TiB	496 TiB	368 TiB x each license
r4.2xlarge	31	368 TiB	368 TiB	496 TiB	368 TiB x each license
r5.xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
r5.2xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license
r5d.2xlarge	19	304 TiB	368 TiB	304 TiB	368 TiB x each license

Aggregate limits

Cloud Volumes ONTAP uses AWS volumes as disks and groups them into *aggregates*. Aggregates provide storage to volumes.

Parameter	Limit
Maximum number of aggregates	Single node: Same as the disk limit HA pairs: 18 in a node [1]
Maximum aggregate size	96 TiB of raw capacity [2]
Disks per aggregate	1-6 [3]
Maximum number of RAID groups per aggregate	1

Notes:

1. It is not possible to create 18 aggregates on both nodes in an HA pair because doing so would exceed the data disk limit.
2. The aggregate capacity limit is based on the disks that comprise the aggregate. The limit does not include object storage used for data tiering.
3. All disks in an aggregate must be the same size.

Logical storage limits

Logical storage	Parameter	Limit
Storage virtual machines (SVMs)	Maximum number for Cloud Volumes ONTAP (HA pair or single node)	One data-serving SVM and one destination SVM used for disaster recovery. You can activate the destination SVM for data access if there's an outage on the source SVM. [1] The one data-serving SVM spans the entire Cloud Volumes ONTAP system (HA pair or single node).
Files	Maximum size	16 TiB
	Maximum per volume	Volume size dependent, up to 2 billion
FlexClone volumes	Hierarchical clone depth [2]	499
FlexVol volumes	Maximum per node	500
	Minimum size	20 MB
	Maximum size	100 TiB
Qtrees	Maximum per FlexVol volume	4,995
Snapshot copies	Maximum per FlexVol volume	1,023

Notes:

1. Cloud Manager does not provide any setup or orchestration support for SVM disaster recovery. It also does not support storage-related tasks on an additional SVM. You must use System Manager or the CLI for SVM disaster recovery.
   - SVM Disaster Recovery Preparation Express Guide
   - SVM Disaster Recovery Express Guide
2. Hierarchical clone depth is the maximum depth of a nested hierarchy of FlexClone volumes that can be created from a single FlexVol volume.

iSCSI storage limits

iSCSI storage	Parameter	Limit
LUNs	Maximum per node	1,024
	Maximum number of LUN maps	1,024
	Maximum size	16 TiB
	Maximum per volume	512
igroups	Maximum per node	256
Initiators	Maximum per node	512
	Maximum per igroup	128
iSCSI sessions	Maximum per node	1,024

iSCSI storage	Parameter	Limit
LIFs	Maximum per port	32
	Maximum per portset	32
Portsets	Maximum per node	256

Storage limits in Azure

Cloud Volumes ONTAP has storage configuration limits to provide reliable operations. For best performance, do not configure your system at the maximum values.

Maximum system capacity by license

The maximum system capacity for a Cloud Volumes ONTAP system is determined by its license. The maximum system capacity includes disk-based storage plus object storage used for data tiering. NetApp doesn't support exceeding this limit.

License	Maximum system capacity (disks + object storage)
Explore	2 TiB (data tiering is not supported with Explore)
Standard	10 TiB
Premium	368 TiB
BYOL	368 TiB per license

For HA, is the license capacity limit per node or for the entire HA pair?

The capacity limit is for the entire HA pair. It is not per node. For example, if you use the Premium license, you can have up to 368 TiB of capacity between both nodes.

Disk and tiering limits by VM size

The disk limits below are specific to disks that contain user data. The limits do not include the boot disk and root disk. The tables below show the maximum system capacity by VM size with disks or alone, and with disks and cold data tiering to object storage.

Disk limits are shown by VM size for Premium and BYOL licenses only because disk limits can't be reached with Explore or Standard licenses due to system capacity limits.

- Single node systems can use Standard HDD Managed Disks, Standard SSD Managed Disks, and Premium SSD Managed Disks, with up to 32 TiB per disk. The number of supported disks varies by VM size.
- HA systems use Premium page blobs as disks, with up to 8 TiB per page blob. The number of supported disks varies by VM size.

Single node with a Premium license

VM size	Max disks per node	Max system capacity with disks alone	Max system capacity with disks and data tiering
DS5_v2	63	368 TiB	368 TiB
DS14_v2	63	368 TiB	368 TiB
DS15_v2	63	368 TiB	368 TiB

Single node with one or more BYOL licenses

For some VM types, you'll need several BYOL licenses to reach the max system capacity listed below. For example, you'd need 6 BYOL licenses to reach 2 PiB with DS5_v2.

VM size	Max disks per node	Max system capacity with one license		Max system capacity with multiple licenses	
		Disks alone	Disks + data tiering	Disks alone	Disks + data tiering
DS3_v2	15	368 TiB	368 TiB	480 TiB	368 TiB x each license
DS4_v2	31	368 TiB	368 TiB	992 TiB	368 TiB x each license
DS5_v2	63	368 TiB	368 TiB	2 PiB	368 TiB x each license
DS13_v2	31	368 TiB	368 TiB	992 TiB	368 TiB x each license
DS14_v2	63	368 TiB	368 TiB	2 PiB	368 TiB x each license
DS15_v2	63	368 TiB	368 TiB	2 PiB	368 TiB x each license

HA pairs with a Premium license

VM size	Max data disks for an HA pair	Max system capacity with disks alone	Max system capacity with disks and data tiering
DS5_v2	63	368 TiB	368 TiB
DS14_v2	63	368 TiB	368 TiB
DS15_v2	63	368 TiB	368 TiB

HA pairs with one or more BYOL licenses

VM size	Max data disks for an HA pair	Max system capacity with one license		Max system capacity with multiple licenses	
		Disks alone	Disks + data tiering	Disks alone	Disks + data tiering
DS4_v2	31	368 TiB	368 TiB	248 TiB	368 TiB x each license
DS5_v2	63	368 TiB	368 TiB	504 TiB	368 TiB x each license
DS13_v2	31	368 TiB	368 TiB	248 TiB	368 TiB x each license
DS14_v2	63	368 TiB	368 TiB	504 TiB	368 TiB x each license
DS15_v2	63	368 TiB	368 TiB	504 TiB	368 TiB x each license

Aggregate limits

Cloud Volumes ONTAP uses Azure storage as disks and groups them into *aggregates*. Aggregates provide storage to volumes.

Parameter	Limit
Maximum number of aggregates	Same as the disk limit
Maximum aggregate size	352 TiB of raw capacity for single node [1, 2] 96 TiB of raw capacity for HA pairs [1]
Disks per aggregate	1-12 [3]
Maximum number of RAID groups per aggregate	1

Notes:

1. The aggregate capacity limit is based on the disks that comprise the aggregate. The limit does not include object storage used for data tiering.
2. The 352 TiB limit is supported starting with 9.6 P3. Releases prior to 9.6 P3 support up to 200 TiB of raw capacity in an aggregate on a single node system.
3. All disks in an aggregate must be the same size.

Logical storage limits

Logical storage	Parameter	Limit
Storage virtual machines (SVMs)	Maximum number for Cloud Volumes ONTAP (HA pair or single node)	One data-serving SVM and one destination SVM used for disaster recovery. You can activate the destination SVM for data access if there's an outage on the source SVM. [1] The one data-serving SVM spans the entire Cloud Volumes ONTAP system (HA pair or single node).

Logical storage	Parameter	Limit
Files	Maximum size	16 TiB
	Maximum per volume	Volume size dependent, up to 2 billion
FlexClone volumes	Hierarchical clone depth [2]	499
FlexVol volumes	Maximum per node	500
	Minimum size	20 MB
	Maximum size	100 TiB
Qtrees	Maximum per FlexVol volume	4,995
Snapshot copies	Maximum per FlexVol volume	1,023

Notes:

1. Cloud Manager does not provide any setup or orchestration support for SVM disaster recovery. It also does not support storage-related tasks on an additional SVM. You must use System Manager or the CLI for SVM disaster recovery.
   - SVM Disaster Recovery Preparation Express Guide
   - SVM Disaster Recovery Express Guide
2. Hierarchical clone depth is the maximum depth of a nested hierarchy of FlexClone volumes that can be created from a single FlexVol volume.

iSCSI storage limits

iSCSI storage	Parameter	Limit
LUNs	Maximum per node	1,024
	Maximum number of LUN maps	1,024
	Maximum size	16 TiB
	Maximum per volume	512
igroups	Maximum per node	256
Initiators	Maximum per node	512
	Maximum per igroup	128
iSCSI sessions	Maximum per node	1,024
LIFs	Maximum per port	32
	Maximum per portset	32
Portsets	Maximum per node	256

Storage limits in Google Cloud

Cloud Volumes ONTAP has storage configuration limits to provide reliable operations. For best performance, do not configure your system at the maximum values.

Maximum system capacity by license

The maximum system capacity for a Cloud Volumes ONTAP system is determined by its license. The maximum system capacity includes disk-based storage plus object storage used for data tiering. NetApp doesn't support exceeding this limit.

For the Premium and BYOL licenses, disk limits prevent you from reaching the 368 TB capacity limit by using disks alone. You can reach the 368 TB capacity limit by tiering inactive data to object storage. Refer to the disk limits below for more details.

License	Maximum system capacity (disks + object storage)
Explore	2 TB (data tiering is not supported with Explore)
Standard	10 TB
Premium	368 TB
BYOL	368 TB per license

Disk and tiering limits

The table below shows the maximum system capacity with disks alone, and with disks and cold data tiering to object storage. The disk limits are specific to disks that contain user data. The limits do not include the boot disk and root disk.

Parameter	Limit
Maximum disks per system	16
Maximum disk size	16 TB
Maximum system capacity with disks alone	64 TB
Maximum system capacity with disks and cold data tiering to a Google Cloud Storage bucket	• Premium: 368 TB • BYOL: 368 TB per license

Aggregate limits

Cloud Volumes ONTAP groups Google Cloud Platform disks into *aggregates*. Aggregates provide storage to volumes.

Parameter	Limit
Maximum number of aggregates	16

Parameter	Limit
Maximum aggregate size	64 TB of raw capacity [1]
Disks per aggregate	1-6 [2]
Maximum number of RAID groups per aggregate	1

Notes:

1. The aggregate capacity limit is based on the disks that comprise the aggregate. The limit does not include object storage used for data tiering.
2. All disks in an aggregate must be the same size.

Logical storage limits

Logical storage	Parameter	Limit
Storage virtual machines (SVMs)	Maximum number for Cloud Volumes ONTAP	One data-serving SVM and one destination SVM used for disaster recovery. You can activate the destination SVM for data access if there's an outage on the source SVM. [1] The one data-serving SVM spans the entire Cloud Volumes ONTAP system.
Files	Maximum size	16 TB
	Maximum per volume	Volume size dependent, up to 2 billion
FlexClone volumes	Hierarchical clone depth [2]	499
FlexVol volumes	Maximum per node	500
	Minimum size	20 MB
	Maximum size	100 TB
Qtrees	Maximum per FlexVol volume	4,995
Snapshot copies	Maximum per FlexVol volume	1,023

Notes:

1. Cloud Manager does not provide any setup or orchestration support for SVM disaster recovery. It also does not support storage-related tasks on an additional SVM. You must use System Manager or the CLI for SVM disaster recovery.
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2. Hierarchical clone depth is the maximum depth of a nested hierarchy of FlexClone volumes that can be created from a single FlexVol volume.

iSCSI storage limits

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	Maximum size	16 TB
	Maximum per volume	512
igroups	Maximum per node	256
Initiators	Maximum per node	512
	Maximum per igroup	128
iSCSI sessions	Maximum per node	1,024
LIFs	Maximum per port	1
	Maximum per portset	32
Portsets	Maximum per node	256

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